Retail Inventory & Sales Analytics Project Documentation

1. Introduction

The Retail Inventory & Sales Analytics Dashboard project was developed to analyze retail business data and convert raw information into meaningful business insights using data analytics tools.

Retail businesses generate large amounts of sales and inventory data daily. Managing this data manually is difficult and time-consuming. This project helps businesses analyze:

- Sales performance
- Inventory status
- Demand forecasting
- Regional performance
- Product category trends

The project demonstrates the complete workflow of a data analytics project from data cleaning to dashboard visualization.

2. Objectives of the Project

The main objective of this project is to help businesses make data-driven decisions and improve operational efficiency through business intelligence and analytics.

1. Analyze Sales Performance:

To identify:

- High performing regions
- Best selling categories
- Overall revenue generation

2. Monitor Inventory Levels:

To identify:

- Low inventory products
- Overstock products
- Inventory risk areas

3. Create Interactive Dashboard:

- To present business insights visually using interactive charts, KPIs, and filters.

4. Improve Business Decision Making:

To help businesses make data-driven decisions related to:

- Inventory management
- Product demand
- Sales strategy

3. Tools & Technologies Used

1. Excel

- Microsoft Excel was used for data preprocessing and cleaning.

Purpose of Using Excel:

Excel was used because raw datasets often contain:

- Missing values
- Duplicate records
- Incorrect formatting
- Data inconsistencies

Data Cleaning Steps Performed in Excel

a) Removed Duplicate Records:

- Duplicate rows were identified and removed to maintain data accuracy.

b) Corrected Date Formats:

- Dates were converted into standard format for proper analysis and visualization.

c) Standardized Column Names:

- Column names were renamed into database-friendly formats such as:
- Store_ID → store_id
- Product_ID → product_id

d) Converted Discount into Percentage Format:

- Discount values were properly formatted into percentage format for calculations.

e) Verified Numeric Columns:

Columns like:

- price
- units_sold
- inventory_level

were checked for correct numeric formats.

2. MySQL

- MySQL was used for database management and SQL analysis.

Purpose of Using MySQL

MySQL was used to:

- Store cleaned data
- Create relational tables
- Perform SQL analysis
- Generate KPIs and business insights

Database Operations Performed

a) Database Creation

A retail analytics database was created to store project data.

b) Table Creation

Fact and dimension tables were created for structured analysis.

Example tables:

- fact_sales
- fact_inventory
- dim_products
- dim_stores

c) CSV File Import

Cleaned CSV files were imported into MySQL tables.

d) SQL Queries

SQL queries were used to:

- Calculate revenue
- Analyze sales
- Detect inventory risk
- Group regional performance

3. Power BI

- Power BI was used to build the interactive dashboard.

Purpose of Using Power BI

Power BI helps convert SQL data into:

- Visual reports
- Interactive charts
- Business dashboards

Dashboard Features

a) KPI Cards:

Used to display:

- Total Revenue
- Total Units Sold
- Average Discount

b) Regional Sales Analysis:

- Bar charts were used to compare sales across regions.

c) Product Category Analysis:

- Pie charts were used to analyze category-wise sales contribution.

d) Inventory Risk Analysis:

- Tables were used to identify products with low inventory levels.

e) Monthly Sales Trend:

- Line charts were used to analyze monthly sales patterns.

f) Interactive Slicers:

Slicers were added for:

- Region
- Category
- Seasonality
- Weather condition

4. Dataset Information

Dataset Description:

- The dataset contains retail sales and inventory data for multiple stores and products.

Columns Included in Dataset

Column Name	**Description**
Date	Transaction date
Store_ID	Unique store identifier
Product_ID	Unique product identifier
Category	Product category
Region	Store region
Inventory_Level	Available stock quantity
Units_Sold	Number of products sold
Units_Ordered	Ordered stock quantity
Demand_Forecast	Predicted future demand
Price	Product price

Column Name	Description
Discount	Discount percentage
Weather_Condition	Weather impact on sales
Holiday/Promotion	Promotional activity indicator
Competitor_Pricing	Competitor product pricing
Seasonality	Seasonal sales pattern

Importance of Dataset

The dataset helps businesses understand:

- Customer demand
- Inventory movement
- Sales performance
- Regional trends
- Seasonal behavior

5. SQL Analysis

- SQL analysis was performed to generate business insights from the dataset.

KPI 1 — Total Revenue Analysis

Objective

- To calculate estimated business revenue.

SQL Logic

Revenue was calculated using:

- units sold
- competitor pricing
- discount

Business Importance

- Helps understand overall business performance.

KPI 2 — Inventory Risk Analysis

Objective

- To identify products with low inventory levels.

SQL Query Used

```
SELECT
   product_id,
   store_id,
   inventory_level
FROM fact_inventory
WHERE inventory_level <= 60;
```

Business Importance

Helps businesses:

- Avoid stock shortages
- Plan restocking
- Maintain product availability

KPI 3 — Overstock Risk Analysis

Objective

- To identify products with excess inventory.

Business Importance

Helps reduce:

- Storage cost
- Inventory wastage
- Unsold stock risk

KPI 4 — Regional Sales Analysis

Objective

- To compare sales performance across regions.

Business Importance

Helps identify:

- Best performing regions
- Low performing regions
- Regional demand patterns

6. Data Modeling

Star Schema Design:

- A star schema model was used for efficient data analysis.

1. Fact Tables

Contain transactional data:

- fact_sales
- fact_inventory

2. Dimension Tables

Contain descriptive information:

- dim_products
- dim_stores

3. Relationships Created

Relationships were created using:

- product_id
- store_id

This enabled Power BI to connect multiple tables for analysis.

7. Power BI Dashboard

Dashboard Components

1. KPI Cards:

Displayed:

- Total Revenue

- Total Units Sold
- Average Discount

2. Regional Sales Bar Chart:

Compared sales performance across:

- North
- South
- East
- West

3. Product Category Pie Chart:

- Displayed category contribution to total sales.

4. Monthly Sales Trend:

- Showed monthly sales growth patterns.

5. Inventory Risk Table:

- Displayed products with low inventory levels.

6. Interactive Slicers:

- Allowed users to filter data dynamically.

8. Key Insights

1. South Region Generated Higher Sales:

- South region showed higher sales due to multiple store contributions.

2. Low Inventory Products Were Identified:

- Some products showed low inventory levels indicating potential stockout risks.

3. Discounts Affected Revenue:

- Higher discounts reduced overall revenue contribution.

4. Category-wise Analysis Improved Understanding:

- Product category analysis helped identify top-performing categories.

5. Monthly Trends Showed Seasonal Patterns:

- Sales trends varied based on seasonality and demand forecasting.

9. Challenges Faced During Project

1. Data Cleaning Issues:

- Incorrect date formats and duplicate records required preprocessing.

2. MySQL Import Errors:

- Large CSV imports and duplicate column issues occurred during import.

3. Relationship Issues in Power BI:

- Improper store-region mapping caused visualization problems.

4. Dashboard Filtering Problems:

- Slicer interaction settings required correction.

10. Conclusion

The Retail Inventory & Sales Analytics Dashboard project successfully demonstrates the end-to-end data analytics workflow using:

- Excel
- MySQL
- Power BI

The project transformed raw retail data into interactive business insights through:

- Data cleaning
- SQL analysis
- Data modeling
- Dashboard visualization

This project helped develop practical skills in:

- Data preprocessing
- SQL querying
- KPI analysis
- Dashboard development
- Business intelligence

The dashboard provides valuable insights into:

- Sales trends
- Inventory management
- Regional performance
- Business decision-making

This project serves as a strong portfolio project for a fresher-level data analyst role.